

Module 3: Data Warehousing & OLAP Cubes

Star/Snowflake Schemas, Data Cube Computation & 5 OLAP Operations

Module III: Data Warehouse & OLAP Technology — Complete 9-Topic Syllabus Tracker

- Topic 21:** Basic Concepts of Data Warehouse & ETL Pipelines

Topic 23: Data Cube Multidimensional Views (Base to Apex Cuboids)

Topic 25: Data Warehouse Design & 4 Architecture Views [UPLOADED PYQ]

Topic 27: Attribute-Oriented Induction (AOI Generalization Algorithm)

Topic 29: Preliminary Warehouse Concepts (Fact vs. Dimension Tables)
- Topic 22:** Data Warehouse Modeling (Star, Snowflake, Fact Constellation)

Topic 24: OLAP Operations (Roll-Up, Drill-Down, Slice, Dice, Pivot)

Topic 26: Data Warehouse Implementation (Materialized Views & Indexing)

Topic 28: Data Cube Computation & Materialization Choices [UPLOADED PYQ]

Topic 21 & 22: Data Warehouse Modeling Schemas [UPLOADED PYQ]

Dimensional Schema	Structural Layout & Normalization	Key Advantages & Tradeoffs
1. Star Schema	Single central Fact Table connected radially to completely denormalized Dimension Tables (1-join query topology).	Simplest structure; blazing fast OLAP query response times. Contains data redundancy in dimension tables.
2. Snowflake Schema [UPLOADED PYQ]	Dimension tables are normalized into multiple hierarchical sub-tables (e.g., `Product` splits into `Product_Category` and `Supplier`).	Zero data redundancy; saves storage space. Slower query execution due to expensive multi-table relational joins.
3. Fact Constellation (Galaxy)	Multiple Fact Tables (e.g., `Sales_Fact` and `Shipping_Fact`) sharing common conformed Dimension Tables (`Time`, `Item`).	Models sophisticated enterprise-wide business operations across multiple departments.

Topic 23 & 24: Data Cube & OLAP Operations [UPLOADED PYQ]

The 5 Canonical OLAP Operations

1. **Roll-Up (Drill-Up):** Aggregates data by climbing up a concept hierarchy or reducing dimensions (e.g., from `Day` → `Month` → `Quarter` → `Year`).
2. **Drill-Down:** Navigates from summarized high-level data to detailed granular data (e.g., from `Country` → `State` → `City`).
3. **Slice:** Performs a selection on a single dimension, producing a 2D sub-cube (e.g., `Time = "Q1 2026"`).
4. **Dice:** Defines a sub-cube by performing a selection on two or more dimensions (e.g., `Location = {"Ranchi", "Patna"}` AND `Item = {"Laptop", "Phone"}`).
5. **Pivot (Rotate):** Rotates the data axes in visualization to provide alternative multidimensional viewing angles.

Feature	OLTP (Online Transaction Processing)	OLAP (Online Analytical Processing) [UPLOADED PYQ]
Primary Focus	Day-to-day transaction processing and CRUD operations.	Complex multidimensional decision support and data mining.
Data Content	Current, real-time snapshot; highly detailed.	Historical, consolidated, multi-year summarized aggregates.
Database Design	Highly normalized entity-relationship (ER) models (3NF).	Denormalized multidimensional Star / Snowflake schemas.
Access Pattern	High frequency, short, deterministic atomic transactions.	Low frequency, complex analytical aggregation queries.

Topic 25 – 28: Warehouse Design Views & Cube Materialization [UPLOADED PYQ]

[UPLOADED PYQ — IT426 Mid] The 4 Essential Design Views for a Data Warehouse

1. **Top-Down View:** Defines the overall information requirements of decision makers and enterprise business goals.
2. **Data Source View:** Models the operational schemas, data types, and access protocols of legacy source systems.
3. **Data Warehouse View:** Defines the dimensional Star/Snowflake schemas, fact measures, dimensions, and aggregation granularities.
4. **Business Query View:** Analyzes user query patterns, analytical workloads, and reporting requirements to guide index design and cube materialization.

Data Cube Materialization Choices (2^n Cuboids) [UPLOADED PYQ]:

- **1. No Materialization:** Compute cuboids on-the-fly when requested. Zero disk overhead; extremely slow query latency.
- **2. Full Materialization:** Precompute all 2^n possible cuboids in the lattice. Fast query response; massive storage overhead.
- **3. Partial Materialization (Optimal Industry Standard):** Precomputes a carefully selected subset of high-frequency cuboids based on query workload and storage constraints (Greedy algorithm / PBS).



M3 Active Recall & Exam Questions [UPLOADED PYQ]

Q1. [UPLOADED PYQ] Compare Star Schema and Snowflake Schema across 4 architectural parameters. (8 Marks)

1. **Normalization Level:** Star schema uses denormalized dimension tables; Snowflake schema normalizes dimension tables into 3NF hierarchies.
2. **Query Performance:** Star schema provides faster query execution because queries require only simple single-level joins with the central fact table; Snowflake requires multiple expensive joins across normalized dimension sub-tables.
3. **Storage Overhead:** Star schema contains data redundancy (e.g., repeated city/state names); Snowflake eliminates redundancy, saving disk space.
4. **Maintenance Complexity:** Star schema is easier to understand and maintain for business analysts; Snowflake has complex structural dependencies.